

June Bang | He/Him

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EDUCATION

University at Buffalo | School of Engineering and Applied Sciences

B.S. Aerospace and Mechanical Engineering + Minor in Robotics

August 2024 – Expected May 2028

EXPERIENCE

University Rover Challenge, University at Buffalo

Chief Engineer

Buffalo, NY

August 2024 – Present

- Led system-level engineering for a 65-member rover team across 9 subteams, owning architecture and interface decisions spanning mechanical, electrical, controls, and software systems
- Reviewed CAD, tolerance stack-ups, GD&T, materials, and manufacturing processes to drive designs from concept through fabrication and assembly
- Coordinated mechanical/software integration using **ROS 2, Gazebo, Isaac Sim, and MoveIt 2**, identifying interface issues before hardware integration

Manufacturing Engineer, Moog

Intern

Buffalo, NY

May 2026 – Present

- Built a standardized Siemens NX CAM tooling library for Nakamura NTRX-300 mill-turn centers, improving programming and setup consistency
- Collaborate directly with machinists and manufacturing engineers to identify tooling and setup issues and incorporate shop-floor feedback into CAD/CAM workflows
- Developed standardized machining documentation to reduce supervised onboarding and prevent setup/tooling errors

SEAS DREAM Lab, University at Buffalo

Student Assistant

Buffalo, NY

September 2025 – Present

- Conducted 6 hands-on workshops on robotics and engineering topics including GD&T, topology optimization, and Arduino-based systems, reaching 5–15 students per session
- Trained 2–3 students per shift on 3D printing and CNC milling operation, supporting safe and effective use of fabrication equipment for rapid prototyping and robotics projects

Projects

Cycloidal Gearbox Design

December 2025 – Present

- Designed and manufactured a 1:64 cycloidal gearbox targeting 40–50 Nm output torque, calculating gear geometry, eccentricity, and an estimated 82% mechanical efficiency
- Tested ~14 arcmin backlash and iterated CF-PETG components, pin clearances, and fits to eliminate load-induced gear skipping

Differential Wrist Joint

August 2024 – May 2025

- Designed a differential wrist mechanism targeting 25 Nm output torque using a 1:3 drive stage and 1:4 differential reduction
- Used FEA, topology optimization, and tolerance analysis to reduce component weight by 36% while maintaining structural integrity and reliable assembly

Skills

Software & Simulation: MATLAB, SolidWorks, Inventor, Siemens NX, AutoCAD, ROS, Gazebo, Python, Java

Fabrication & Prototyping: 3D Printing, CNC Milling, Machining, CAM, Composite Materials

Analysis & Modeling: Finite Element Analysis (FEA), GD&T, Topology Optimization, Static Load Analysis